K. J. SOMAIYA INSTITUTE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

Aviation Weather Forecast Dashboard

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Abstract

Aviation weather forecasts play a critical role in ensuring flight safety and operational efficiency. This project presents a web-based Aviation Forecast Verification Dashboard developed to automate the verification of aviation weather forecasts issued by the India Meteorological Department (IMD). The system processes forecast files, compares them with actual METAR and upper-air observations, and evaluates forecast accuracy through automated parameter matching and time-based verification. Additionally, a Machine Learning module uses Logistic Regression to predict potential thunderstorm and strong-wind events, while Linear Regression estimates wind speed, gust and direction for the next 25–40 minutes.

Background

- Aviation operations depend on accurate and timely weather forecasts.
- Manual verification of aviation forecasts involves processing multiple files and comparing observations, which is time-consuming.
- Forecast data and actual weather observations often require different formats and time references.
- Early identification of potential thunderstorms and strong winds can support better operational awareness.



Figure 1: Manual Aviation Forecast Verification Process.

Objectives

- Automate the verification of Take-off, Local Area and Aerodrome Weather Forecasts.
- Compare forecast parameters with actual METAR and upper-air observations.
- Calculate forecast accuracy and generate verification reports.
- Automate time matching, parameter extraction and warning validation.
- Predict potential thunderstorm (TSRA) and strong-wind/gust events using Machine Learning.
- Forecast surface wind speed, maximum gust and wind direction.
- Provide a centralized dashboard for analysis and reporting.

Methodology

1. **Data Collection:** Forecast files, METAR and upper-air observations.
2. **Data Processing:** Parsing, time matching and parameter extraction.
3. **Verification:** Compare forecasts with actual observations and calculate accuracy.
4. **ML Prediction:** Logistic Regression for hazard classification and Linear Regression for wind forecasting.
5. **Output:** Dashboard, warning predictions, graphs and reports.

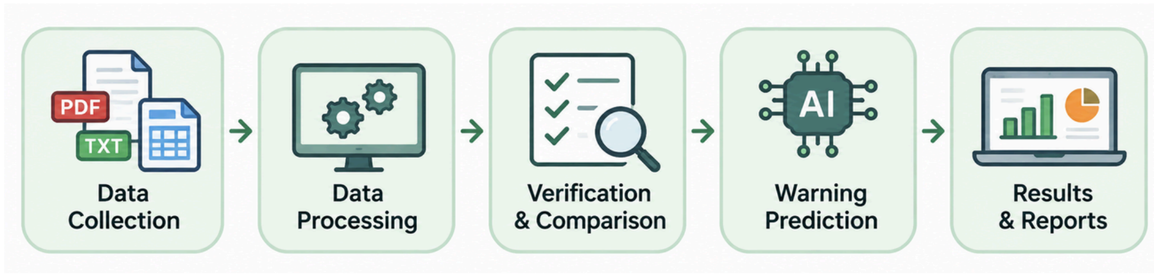


Figure 3: Proposed Methodology for Aviation Forecast Verification and AI-Based Warning Prediction.

Result and Discussion

- Automated forecast verification and accuracy reporting.
- Comparison of forecast and observed parameters.
- Prediction of TSRA and strong-wind/gust events.
- Short-term wind speed, gust and direction forecasts.

Overall, the system demonstrates strong real-time performance and usability.

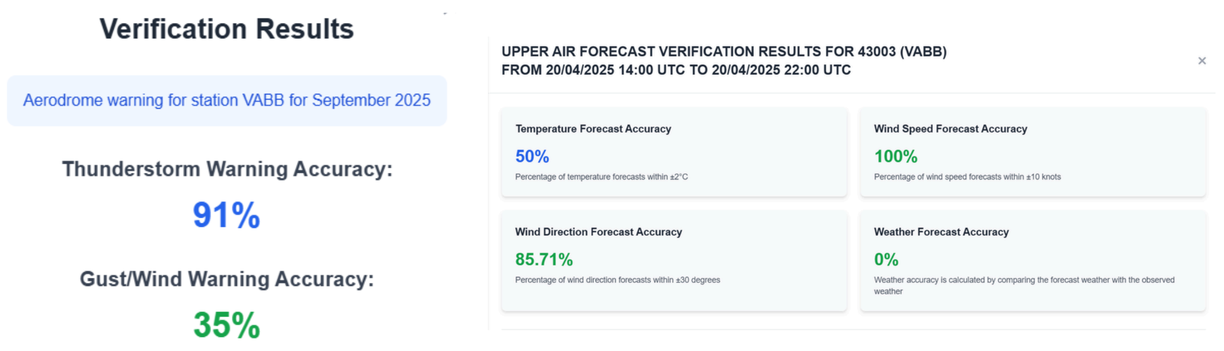


Figure 3: Dashboard Outputs – Forecast Verification, ML-Based Warning Prediction and Results & Reports.

Conclusion

The system combines automated forecast verification with ML-based hazard prediction, reducing manual effort and supporting improved aviation weather analysis.

Achievements

- Received an internship certificate from IMD for project development.
- Successfully implemented the dashboard at an IMD station in Maharashtra.
- Currently undergoing testing and validation.
- Additional features are under development, with the long-term goal of expanding deployment across India (Pan-India).

Acknowledgement

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